

KARTIK MISHRA

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EDUCATION

Amity University, Noida, India
B.Tech. Biotechnology

Aug 2017 – Mar 2021
CGPA: 8.15/10

WORK EXPERIENCE

BHARTI AIRTEL

Machine Learning Engineer

Gurugram, India

Aug 2025 – Present

- Designed and built a real-time conversational voice bot that helps field engineers resolve broadband complaints at customer premises – architecting Meta's OmniASR (STT), Qwen3.5 (LLM) and Voxtral (TTS) turn-taking loop in LangGraph across decision, agent, and communication nodes, achieving sub-1,000 ms round-trip latency. Built the asynchronous FastAPI microservices and a MongoDB-backed priority call queue and state machine to manage concurrent live calls over Airtel IQ WebSocket telephony.
- Finetuned Meta's Omnilingual ASR (OmniASR 1B/3B) and Whisper-Apex for Hindi/Hinglish speech, engineering a μ -law + noise augmentation pipeline that injected real noise profiles mined from in-house customer call recordings into clean training data to match production telephony conditions; reached ~18.9% WER / ~9.0% UER on a held-out validation set.
- Engineered the ASR serving infrastructure across two lanes – OmniASR (wav2vec2 + LLM-prefix) on a dedicated Nvidia's Triton lane and Whisper-Apex on vLLM – with a real-time audio front-end using Silero VAD for speech segmentation and DeepFilterNet noise suppression, plus bandwidth-extension stages to close a confirmed 8 – 16 kHz telephony audio ceiling.
- Revamped the WCL stack (2G/4G/5G) and built the team's first Docker-based Apache Airflow, CI/CD-enabled orchestration – migrating pipelines between PySpark and pandas by workload profile and orchestrating 20+ ETL workflows as SME, cutting runtime 75% and compute cost 85%.
- Leading nationwide Duct operations across 12+ states – a business-critical program directly affecting customer network performance, traffic, and tower/cell operations – as tech lead and primary SME/POC coordinating cross-functional teams India-wide; deployed a real-time monitoring dashboard that eliminated manual inspections and gave stakeholders instant operational visibility.
- Built large-scale PySpark pipelines processing terabytes of broadband customer data daily across India, generating network-health KPIs, churn signals, and CX metrics for Airtel's flagship Broadband Experience initiative.

MOODY'S INSURANCE SOLUTIONS

Associate Analytics, Model Analyst

Noida, India

Apr 2023 – Aug 2025

- Architected an end-to-end data orchestration framework on Dagster, integrating high-frequency weather data from 30+ global sources with CI/CD automation via Jenkins and GitHub Actions, improving pipeline efficiency by 90%.
- Implemented a Django-based data visualization dashboard using Mapbox, JavaScript, MongoDB, and Python, enabling global wind station tracking and cross-referencing with the Postgres database to identify over 18,000 missing stations
- Collaborated with the Event Response team to automate Support Center uploads via an API, reducing manual upload time from 40–60 minutes to zero, enhancing efficiency for natural disaster reports and forecasts
- Unlocked Visibility into HWind processes by building insightful statistics dashboard using JavaScript, Python, and Django, tracking daily and weekly data feed counts with interactive charts, identifying any problematic data transactions and feed downtimes.

VERISK ANALYTICS (WOOD MACKENZIE)

Data Associate

Gurugram, India

Nov 2021 – Apr 2023

- Enhanced data collection and processing pipelines in Python on AWS SageMaker, improving efficiency by 60%
- Built Python automation scripts to extract oil price data from country reports, update Excel deliverables, and trigger email notifications on completion, reducing end-to-end processing time by 90%.
- Led project handover sessions with different teams, managed project roadmaps, story allocation, and task division using JIRA
- Crafted a Power BI dashboard to monitor JIRA tickets and track completion statistics, fostering team accountability and ensuring timely task fulfillment; implemented JIRA tickets for day-to-day tasks assigned to individual team members

PUBLICATIONS

- Identification of vaccine targets & design of vaccine against SARS-CoV-2 coronavirus using computational and deep learning-based approaches - PeerJ, May 2022
- Identification of vaccine targets in pathogens and design of a vaccine using computational approaches - Nature Scientific Reports, September 2021
- To Study the Effect of Unconventional Treatment Protocol on COVID-19 Patients in Delhi Using Artificial Intelligence Based Methods - SSRN, September 2020

EXTRACURRICULAR ACTIVITIES

Member & AVP Education, Toastmasters Club, West Delhi – Organized weekly meetings, speaker selection, and onboarded new members (Apr 2023 – Mar 2025)

Volunteer & Drive Lead, Robin Hood Army, Delhi – Led 30+ food distribution drives and taught 25+ students weekly (Oct 2022 – Present)

Coordinator, CRC Placement Cell, Amity University – Orchestrated 100+ on-campus placement drives and pre-placement prep sessions (Feb 2019 – Mar 2021)

SKILLS

Languages: Python, SQL

Big Data: Apache PySpark, Pandas, Hadoop, HDFS, YARN, Delta Lake

Orchestration: Apache Airflow, Docker, Docker Compose, Dagster, Jenkins, GitHub Actions, Bitbucket

ML/AI: PyTorch, Transformers, HuggingFace, LangGraph, LangChain, LLMs, Speech-to-Text/Text-to-Speech (STT/TTS), ASR finetuning, OmniASR, Whisper, DeepFilterNet, Silero VAD, noise suppression

Backend & Web: FastAPI, Streamlit, WebSockets, Django

Databases: PostgreSQL, MongoDB

Visualization: Django, Power BI, Tableau

Tools: Power BI, Tableau, Git, Linux, Paramiko, AWS SageMaker

Spoken: English (Proficient), Hindi (Native)